HW12

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Question(s):

Write a paper review for *DEEPSERVE: Serverless Large Language Model Serving at Scale*Links to an external site., following the format below:

Overall score (1: reject, 2: weak reject, 3: major revision, 4: minor revision, 5: accept)

Summary of the paper (be concise, in one paragraph)

Reasons for accepting the paper (list in a pointed format, i.e., "- reason1, - reason 2")

Reasons for rejecting the paper (the same as the reasons for acceptance)

Comments for the authors (the detailed opinions on this paper, including explaining your score, asking questions, suggesting improvements, etc.)

Answers:

1. Overall score

Minor revision

2. Summary of the paper

The paper presents DEEPSERVE, Huawei Cloud's production serverless platform for LLM workloads built on Ascend NPUs. It introduces a request–job–task abstraction for unifying fine-tuning, agent, and model serving; a microkernel-style, NPU-centric serving engine (FLOWSERVE) with SPMD execution and a Relational Tensor Cache plus DistFlow for KV/tensor movement; distributed schedulers that handle locality and prefill/decode (PD) disaggregation; and a set of scaling optimizations (pre-warmed pods/TEs, DRAM preloading, NPU-fork) that let the system scale to 64 instances in seconds and have been used in production for over a year.

3. Reasons for accepting the paper

- Production system with a full platform (APIs, serverless abstraction, cluster manager, engine) on large Ascend NPU clusters.
- Well-structured engine design (microkernel, NPU-centric, SPMD, centralized async scheduler, RTC + DistFlow) that aligns with and extends current LLM-serving practice.
- Concrete scheduling and scaling techniques (locality + PD-aware scheduling, pre-warmed pods/TEs, DRAM pre-load, NPU-fork) with clear latency and scale-up improvements.
- Broad coverage of workloads (fine-tuning, agents, dense/MoE serving) and hardware generations (scale-out clusters + SuperPod) that makes the paper practically useful.

4. Reasons for rejecting the paper

- Limited external comparison: Evaluation mainly uses internal baselines and hardware, with no direct comparison to state-of-the-art LLM engines/serverless systems (e.g., vLLM, SGLang, ServerlessLLM, DistServe, Llumnix), making it hard to quantify relative gains.
- "Serverless" side under-quantified: The paper gives little data on serverless overheads, cold-start behavior vs conventional autoscaling, or multi-tenant isolation beyond the core serving metrics.

5. Comments for the authors

I am not expert in this, but this is a strong, production-driven system paper. The some improvements can be done might be to include one quantitative comparison or careful case study against an existing LLM engine.